Group II, Row IV: reduction of the master expression to the three-term ($\mathcal{P}_{gg}$) form

Claim. The master expression

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w',x'} e^{-ik(w'-w)}\int_{x,y,y'}\frac{(w-z)^m\,(x-y)^k\,(w'-z)^{m'}\,(x'-y')^{k'}}{(w-z)^2\,(x-y)^2\,(w'-z)^2\,(x'-y')^2}\,\frac{p^+\,k^+}{(p^++k^+)^4}\\
&\quad \delta^{(2)}\left[y-w+\frac{p^+}{(p^++k^+)}\,(w-z)\right]\,\delta^{(2)}\left[y'-w'+\frac{p^+}{(p^++k^+)}\,(w'-z)\right]\times\\
&\quad \left[\delta_{km}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{kj}\right]\left[\delta_{k'm'}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm'}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im'}\,\delta_{k'j}\right]\\
&\quad \left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\left[f^{cda}\,U^{ce}(y)\,U^{db}(z)\,\rho^{e}(x)-f^{cda}\,U^{db}(z)\,U^{ce}(x)\,\rho^{e}(x)\right]\end{aligned}\tag{1}$$

is identically equal, up to $O(\Lambda)$ and $O(\epsilon)\times$ finite, to the three-term form

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\\
&\quad \left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\right]\right]\times\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\\
&\quad +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\left[\log\frac{(\vee-k^+)\,k^+}{\vee\,\Lambda}\right]\int_{x,y,x',y'}e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\quad \left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\\
&\quad +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^++\Lambda}}d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'}e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right.\\
&\quad \times\left[U^{ce'}(y')-U^{ce'}(x')\right]\,[U^{ce}(y)-U^{ce}(x)]\,\rho^{e'}(x')\rho^e(x)\,,\end{aligned}\tag{2}$$

where in the first term $\frac{\xi}{(1-\xi)_+}$ denotes the *locally subtracted* prescription, $\int_{k^+/\vee}^1 d\xi\,\frac{h(\xi)}{(1-\xi)_+}\equiv\int_{k^+/\vee}^1 d\xi\,\frac{h(\xi)-h(1)}{1-\xi}$, whose extracted endpoint is the second (blue) term.

1 Colour algebra

The two colour brackets factor as $f^{c'd'a}U^{d'b}(z)[U^{c'e'}(y')-U^{c'e'}(x')]\rho^{e'}(x')$ and $f^{cda}U^{db}(z)[U^{ce}(y)-U^{ce}(x)]\rho^e(x)$. Using the reality/orthogonality of the adjoint Wilson line, $U^{d'b}(z)U^{db}(z)=\delta^{d'd}$, and then $f^{c'da}f^{cda}=N_c\,\delta^{c'c}$,

$$\text{(colour brackets)}\;=\;N_c\left[U^{ce'}(y')-U^{ce'}(x')\right]\left[U^{ce}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^e(x)\,.\tag{3}$$

2 δ-function integration over w, w'

With $\lambda=\frac{p^+}{p^++k^+}$ and $1-\lambda=\frac{k^+}{p^++k^+}$,

$$\delta^{(2)}\big[y-(1-\lambda)w-\lambda z\big]=\left(\frac{p^++k^+}{k^+}\right)^2\delta^{(2)}\left[w-\frac{(p^++k^+)y-p^+z}{k^+}\right],\tag{4}$$

and at the solution

$$w-z=\frac{p^++k^+}{k^+}\,(y-z)\,,\qquad w'-w=\frac{p^++k^+}{k^+}\,(y'-y)\,.\tag{5}$$

Hence, per side, $\int d^2w\,\delta^{(2)}\,\frac{(w-z)^m}{(w-z)^2} = \frac{p^+{+}k^+}{k^+}\,\frac{(y-z)^m}{(y-z)^2}$, the phase becomes $e^{-ik(y'-y)\frac{p^+{+}k^+}{k^+}}$, and the prefactor collects to

$$\frac{p^+k^+}{(p^+{+}k^+)^4}\Big(\frac{p^+{+}k^+}{k^+}\Big)^2 = \frac{p^+}{k^+\,(p^+{+}k^+)^2}\,. \tag{6}$$

3 Tensor contraction over i, j

With $A = \frac{p^+{+}k^+}{p^+}$, $B = \frac{p^+{+}k^+}{k^+}$, a direct index computation (checked mechanically over all index values) gives

$$\sum_{i,j} T_1^{(km,ij)}T_2^{(k'm',ij)} = \big[d_\perp - 2(A+B)\big]\,\delta_{km}\delta_{k'm'} + (A^2+B^2)\,\delta_{mm'}\delta_{kk'} + 2AB\,\delta_{mk'}\delta_{km'}\,, \tag{7}$$

and multiplying by $\frac{p^+}{k^+(p^+{+}k^+)^2}$,

$$-2(A+B) \to -\frac{2}{k^{+2}}\,, \qquad A^2+B^2 \to \frac{1}{k^+}\Big(\frac{1}{p^+}+\frac{p^+}{k^{+2}}\Big)\,, \qquad 2AB \to +\frac{2}{k^{+2}}\,, \qquad d_\perp \to \frac{d_\perp\,p^+}{k^+(p^+{+}k^+)^2}\,. \tag{8}$$

Since the z-integral below is a *symmetric* tensor in (m, m'), the $\delta_{km}\delta_{k'm'}$ and $\delta_{mk'}\delta_{km'}$ channels contract identically with the kernels, so **the $\mp\frac{2}{k^{+2}}$ pieces cancel exactly**: no antisymmetric channel survives.

4 The z integral

Only $\mathcal{K}^m(y-z)\mathcal{K}^{m'}(y'-z)$ depends on z. In $d_\perp = 2-\epsilon$ ($\overline{\rm MS}$), with $r = y'-y$,

$$\int_z \frac{(y-z)^m\,(y'-z)^{m'}}{(y-z)^2\,(y'-z)^2} = -\frac{\pi}{2}\,\underbrace{\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu^2_{\overline{MS}}}{4}\Big]}_{L_\epsilon}\,\delta^{mm'} \;-\;\pi\,\frac{r^m r^{m'}}{r^2}\,. \tag{9}$$

Independent check: the traceless projection of the left-hand side is finite in $d=2$ and was evaluated numerically: it equals $-\pi\big[\frac{r^m r^{m'}}{r^2} - \frac{\delta^{mm'}}{2}\big]$ to 10^{-6}; the trace grows with the infrared cutoff with the r-dependence $-\pi\log r^2$, fixing the structure of (9) (the constant defines $\mu_{\overline{MS}}$). Contracting with the surviving channels of Section 3 and with $\mathcal{K}^k(x-y)\mathcal{K}^{k'}(x'-y')$:

- channel $\frac{d_\perp p^+}{k^+(p^+{+}k^+)^2}\,\delta_{km}\delta_{k'm'}$: $\;-\frac{\pi}{2}L_\epsilon\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y') - \pi\,\frac{[\mathcal{K}(x-y)\cdot r][\mathcal{K}(x'-y')\cdot r]}{r^2}$ (the azimuthal kernel, carrying $d_\perp$);
- channel $\frac{1}{k^+}\big(\frac{1}{p^+}+\frac{p^+}{k^{+2}}\big)\,\delta_{mm'}\delta_{kk'}$: trace of (9) times the dipole, $\big[-\frac{\pi}{2}d_\perp L_\epsilon - \pi\big]\,\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')$, where the extra $-\pi$ comes from $\delta_{mm'}\frac{r^m r^{m'}}{r^2} = 1$. **This $-\pi$ is the origin of the "-1" in the bracket of** (2)**.**

5 ξ substitution and matching

Set $\xi = \frac{k^+}{p^+{+}k^+}$, so that

$$p^+ = \frac{k^+(1-\xi)}{\xi}, \quad dp^+ = \frac{k^+ d\xi}{\xi^2}, \quad e^{-ik(y'-y)\frac{p^+{+}k^+}{k^+}} = e^{-ik(y'-y)/\xi}, \quad p^+ \in (\Lambda, \vee - k^+) \leftrightarrow \xi \in \Big(\frac{k^+}{\vee}, \frac{k^+}{k^+{+}\Lambda}\Big), \tag{10}$$

$$\frac{p^+}{(p^+{+}k^+)^2} = \frac{\xi(1-\xi)}{k^+}, \qquad \frac{p^+}{k^{+2}} = \frac{1-\xi}{\xi k^+}, \qquad \frac{1}{p^+} = \frac{\xi}{(1-\xi)k^+}\,. \tag{11}$$

The L_ϵ coefficient assembles into $\frac{d_\perp}{2}\,\hat p(\xi)$ with $\hat p(\xi) = \xi(1-\xi) + \frac{1-\xi}{\xi} + \frac{\xi}{1-\xi}$, and the overall constant is $\frac{1}{(2\pi)^3}\frac{g^4 N_c}{8\pi^4}\frac{1}{k^+} = \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}$, as in (2). Altogether,

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\,C\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+{+}\Lambda}} d\xi\,\Big\{\frac{d_\perp}{2}\,\frac{\hat p(\xi)}{\xi^2}\,L_\epsilon + \frac{1}{\xi^2}\Big[\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big]\Big\}\,\mathcal{K}\cdot\mathcal{K}'\,f(\xi) \;-\; C\,d_\perp\!\int d\xi\,\frac{1-\xi}{\xi}\,\frac{(\mathcal{K}\cdot r)(\mathcal{K}'\cdot r)}{r^2}\,f(\xi)\,, \tag{12}$$

with $C = \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}$, $f(\xi) = e^{-ik(y'-y)/\xi}$, $\mathcal{K} = \mathcal{K}(x-y)$, $\mathcal{K}' = \mathcal{K}(x'-y')$, the colour brackets understood. Two exact regroupings turn (12) into (2):

(a) The endpoint. For the $\frac{\xi}{1-\xi}$ channel, with $h(\xi)$ the rest of the integrand,

$$\int_{\frac{k^+}{\vee}}^{\frac{k^+}{k^+ + \Lambda}} d\xi \, \frac{h(\xi)}{1-\xi} = \int_{\frac{k^+}{\vee}}^{1} d\xi \, \frac{h(\xi) - h(1)}{1-\xi} \; + \; h(1) \, \log \frac{(\vee - k^+)\, k^+}{\vee \Lambda} \; + \; O(\Lambda)\,, \tag{13}$$

which is exactly the locally subtracted $\frac{\xi}{(1-\xi)_+}$ of the first term of (2) plus the blue endpoint term, both carrying the same bracket $-\big[\frac{d_\perp}{2} L_\epsilon + 1\big]$.

(b) The "-1" and the azimuthal $\mathcal{K}\cdot\mathcal{K}'$ piece. The target writes the L_ϵ-free dipole piece as "-1" against the *full* $\hat{p}/\xi^2$ plus the $+\mathcal{K}\cdot\mathcal{K}' \, \frac{1-\xi}{\xi}$ term inside the third line of (2). This is identical to the L_ϵ-free piece of (12) because of the exact identity

$$\frac{\hat{p}(\xi)}{\xi^2} \; - \; \frac{1-\xi}{\xi} \; = \; \frac{1}{\xi^2}\Big[\frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}\Big]\,. \tag{14}$$

Finally, the r-tensor part of (12) carries exactly $-\, d_\perp \frac{1-\xi}{\xi}$, matching the second kernel of the third term of (2).

Conclusion

Equations (1) and (2) are identical, up to $O(\Lambda)$ (from the endpoint recombination and from extending regular ξ-integrals to $\xi = 1$) and $O(\epsilon) \times$ finite. All coefficients check, including: the overall N_c; the "-1" in the dimensional-regularization bracket (the trace of the r-tensor part of the z-integral against the $\frac{1}{p^+} + \frac{p^+}{k^{+2}}$ channel); the $d_\perp$ on the azimuthal kernel; and the complete cancellation of the antisymmetric $\frac{2}{k^+}(\delta_{mk'}\delta_{m'k} - \delta_{m'k'}\delta_{km})$ channel between the two crossed contractions (the z-integral is symmetric in m, m').

Convention remark. The index algebra is carried out in $d_\perp = 2 - \epsilon$ while the position-space measures and the δ-functions are kept two-dimensional; only the z loop is dimensionally continued. This is consistent as long as it is applied uniformly — as it is on both sides of the equality — since any $\xi^{2\epsilon}$ Jacobian refinement would only shift scheme-dependent finite pieces that cancel in the μ-independent total.